

lyche-numerical-linear-algebra — formula report

4638 inline formulas (section omitted; all rendered), 1056 display equations, 7 tables, 58 unrecovered image regions.

Corrected (8)

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
lyche-numerical-linear-algebra- E08326 (was)	133	0.118	$\begin{aligned} & \boldsymbol{Q}_{\{1\}}:=\left[\begin{array}{lllll} \boldsymbol{q}_{\{1\}}, & \ldots, & \boldsymbol{q}_{\{n\}} \end{array} \right], & \text{and} \\ & \boldsymbol{R}_{\{1\}}:= \left[\begin{array}{ccccc} \ \boldsymbol{v}_1\ _2 & \boldsymbol{a}_2^T \boldsymbol{q}_1 & \boldsymbol{a}_3^T \boldsymbol{q}_1 & \cdots & \boldsymbol{a}_{n-1}^T \boldsymbol{q}_1 \\ 0 & \ \boldsymbol{v}_2\ _2 & \boldsymbol{a}_3^T \boldsymbol{q}_2 & \cdots & \boldsymbol{a}_{n-1}^T \boldsymbol{q}_2 \\ & 0 & \ \boldsymbol{v}_3\ _2 & \cdots & \boldsymbol{a}_{n-1}^T \boldsymbol{q}_2 \\ & & \ddots & \ddots & \boldsymbol{a}_n^T \boldsymbol{q}_3 \\ & & & \ddots & \vdots \\ & & & \ \boldsymbol{v}_{n-1}\ _2 & \boldsymbol{a}_n^T \boldsymbol{q}_{n-1} \\ & & & 0 & \ \boldsymbol{v}_n\ _2 \end{array} \right]. \end{aligned}$	—	
lyche-numerical-linear-algebra- E08326 (now)	133	—	$\begin{aligned} & \boldsymbol{Q}_{\{1\}}:=\left[\begin{array}{l} \boldsymbol{q}_{\{1\}}, \ldots, \boldsymbol{q}_{\{n\}} \end{array} \right], \\ & \text{and} \\ & \boldsymbol{R}_{\{1\}}:= \left[\begin{array}{ccccc} \ \boldsymbol{v}_1\ _2 & \boldsymbol{a}_2^T \boldsymbol{q}_1 & \boldsymbol{a}_3^T \boldsymbol{q}_1 & \cdots & \boldsymbol{a}_{n-1}^T \boldsymbol{q}_1 \\ 0 & \ \boldsymbol{v}_2\ _2 & \boldsymbol{a}_3^T \boldsymbol{q}_2 & \cdots & \boldsymbol{a}_{n-1}^T \boldsymbol{q}_2 \\ & 0 & \ \boldsymbol{v}_3\ _2 & \cdots & \boldsymbol{a}_{n-1}^T \boldsymbol{q}_3 \\ & & \ddots & \ddots & \vdots \\ & & & \ddots & \ \boldsymbol{v}_{n-1}\ _2 \\ & & & & 0 \end{array} \right]. \end{aligned}$	—	
basis: inferred / ink					
Conf. MathPix confidence: ≥0.9 0.5–0.9 <0.5 — no value					
Residual render vs scan (inkdrill): C component W weak S stable N noise K clean not measured					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
lyche-numerical-linear-algebra-EQ0369 (was)	150	■0.298	$\boldsymbol{J}:=\operatorname{diag}\left(\boldsymbol{U}_1,\boldsymbol{U}_2\right)=\left[\begin{array}{cccccc} 2 & 1 & 0 & & & \\ 0 & 2 & 1 & & & \\ 0 & 0 & 2 & & 1 & \\ & & 0 & 1 & & \\ & & 0 & 2 & & \\ & & & & 0 & \\ & & & & 0 & 1 \end{array}\right] \in \mathbb{R}^{8 \times 8}.$	—	
lyche-numerical-linear-algebra-EQ0369 (now)	150	—	$d_i \text{ for } i = 1, \ldots, n, b_{i+1,i} := a_i, b_{i+1} := c_i \text{ for } i \text{ otherwise. } \in \mathbb{C}^{m \times n} \text{ and } 1 \leq i_1 < i_2 < \cdots < i_r \leq m, 1 \leq j \text{ matrix } A(i,j) \in \mathbb{C}^{r \times c} \text{ is the submatrix of } A$	—	
basis: inferred / ink					
lyche-numerical-linear-algebra-EQ0579 (was)	215	■0.322	$\begin{array}{l} \boldsymbol{A}^* \boldsymbol{A} \boldsymbol{x} = \left[\begin{array}{ccc} 1 & \cdots & 1 \\ t_1 & \cdots & t_m \end{array} \right] \left[\begin{array}{c} 1 \\ \vdots \\ 1 \\ 1 \\ m \end{array} \right] \left[\begin{array}{c} x_1 \\ x_2 \end{array} \right] = \left[\begin{array}{cc} m & \sum t_k \\ \sum t_k & \sum t_k^2 \end{array} \right] \left[\begin{array}{c} x_1 \\ x_2 \end{array} \right], \\ = \left[\begin{array}{ccc} 1 & \cdots & 1 \\ t_1 & \cdots & t_m \end{array} \right] \left[\begin{array}{c} y_1 \\ \vdots \\ y_m \end{array} \right] = \left[\begin{array}{c} \sum y_k \\ \sum t_k y_k \end{array} \right] = \boldsymbol{A}^* \boldsymbol{b}, \end{array}$	—	
lyche-numerical-linear-algebra-EQ0579 (now)	215	—	$\text{nal if } a_{ij} = 0 \text{ for } i \neq j. \text{ triangular or right triangular if } a_{ij} = 0 \text{ for } i > j. \text{ triangular or left triangular if } a_{ij} = 0 \text{ for } i < j. \text{ Hessenberg if } a_{ij} = 0 \text{ for } i > j + 1. \text{ Hessenberg if } a_{ij} = 0 \text{ for } i < j + 1. \text{ gonal if } a_{ij} = 0 \text{ for } i - j > 1. \text{ ded if } a_{ij} = 0 \text{ for } i - j > d. \text{ the following notations for diagonal- and tridiagonal } n \times n$	(not rendered)	—
basis: inferred / ink					
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5-0.9 ■ <0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
lyche-numerical-linear-algebra-EQ0630 (was)	229	0.167	$\begin{aligned} & \left[\begin{array}{cc} \boldsymbol{0} & \boldsymbol{A} \\ \boldsymbol{A}^* & \boldsymbol{0} \end{array} \right] \left[\begin{array}{c} \boldsymbol{u}_i \\ \boldsymbol{v}_i \end{array} \right] = \left[\begin{array}{c} \boldsymbol{A} \boldsymbol{v}_i \\ \boldsymbol{A}^* \boldsymbol{u}_i \end{array} \right] = \left[\begin{array}{c} \alpha_i \boldsymbol{u}_i \\ \alpha_i \boldsymbol{v}_i \end{array} \right] = \alpha_i \left[\begin{array}{c} \boldsymbol{u}_i \\ \boldsymbol{v}_i \end{array} \right], \quad i = 1, \dots, r, \\ & \left[\begin{array}{cc} \boldsymbol{0} & \boldsymbol{A} \\ \boldsymbol{A}^* & \boldsymbol{0} \end{array} \right] \left[\begin{array}{c} \boldsymbol{u}_i \\ -\boldsymbol{v}_i \end{array} \right] = \left[\begin{array}{c} -\boldsymbol{A} \boldsymbol{v}_i \\ \boldsymbol{A}^* \boldsymbol{u}_i \end{array} \right] = \left[\begin{array}{c} -\alpha_i \boldsymbol{u}_i \\ \alpha_i \boldsymbol{v}_i \end{array} \right] = -\alpha_i \left[\begin{array}{c} \boldsymbol{u}_i \\ -\boldsymbol{v}_i \end{array} \right], \quad i = 1, \dots, r, \\ & \left[\begin{array}{cc} \boldsymbol{0} & \boldsymbol{A} \\ \boldsymbol{A}^* & \boldsymbol{0} \end{array} \right] \left[\begin{array}{c} \boldsymbol{0} \\ \boldsymbol{0} \end{array} \right] = \left[\begin{array}{c} \boldsymbol{0} \\ \boldsymbol{A}^* \boldsymbol{u}_i \end{array} \right] = \left[\begin{array}{c} \boldsymbol{0} \\ \boldsymbol{0} \end{array} \right] = 0 \left[\begin{array}{c} \boldsymbol{u}_i \\ \boldsymbol{0} \end{array} \right], \quad i = r + 1, \dots, m, \\ & \left[\begin{array}{cc} \boldsymbol{0} & \boldsymbol{A} \\ \boldsymbol{A}^* & \boldsymbol{0} \end{array} \right] \left[\begin{array}{c} \boldsymbol{0} \\ \boldsymbol{v}_i \end{array} \right] = \left[\begin{array}{c} \boldsymbol{A} \boldsymbol{v}_i \\ \boldsymbol{0} \end{array} \right] = \left[\begin{array}{c} \boldsymbol{0} \\ \boldsymbol{0} \end{array} \right] = 0 \left[\begin{array}{c} \boldsymbol{0} \\ \boldsymbol{v}_i \end{array} \right], \quad i = r + 1, \dots, n \end{aligned}$		
lyche-numerical-linear-algebra-EQ0630 (now)	229	—	$\begin{aligned} & \mathbf{a}_{11}x_1 + \mathbf{a}_{12}x_2 + \cdots + \mathbf{a}_{1n}x_n = b_1 \\ & \mathbf{a}_{21}x_1 + \mathbf{a}_{22}x_2 + \cdots + \mathbf{a}_{2n}x_n = b_2 \\ & \quad \vdots \quad \quad \quad \vdots \quad \quad \quad \vdots \\ & \mathbf{a}_{m1}x_1 + \mathbf{a}_{m2}x_2 + \cdots + \mathbf{a}_{mn}x_n = b_m \end{aligned}$		
basis: inferred / ink					
lyche-numerical-linear-algebra-EQ0670 (was)	241	0.390	$\begin{aligned} & \mathbf{a}_{i\ i} \ \&=2\ d, \ i=1, \ldots, n, \ \backslash\ \mathbf{a}_{i, \ i+1}=\mathbf{a}_{i+1, \ i} \ \&=a, \ \text{quad } i=1, \ldots, n-1, \ \text{quad } i \not= m, \ 2\ m, \ \ldots, (m-1)\ m, \\ & \not= m, \ 2\ m, \ \ldots, (m-1)\ m, \ \backslash\ \mathbf{a}_{i, \ i+m}=\mathbf{a}_{i+m, \ i} \ \&=a, \ \text{quad } i=1, \ldots, n-m, \ \backslash\ \mathbf{a}_{i\ j} \ \&=0, \ \text{text } \{ \text{otherwise} \} \end{aligned}$	$\begin{aligned} & a_{ii} = 2d, i = 1, \dots, n, \\ & a_{i,i+1} = a_{i+1,i} = a, \quad i = 1, \dots, n - 1, \quad i \neq m, 2m, \dots, (m - 1)m, \\ & a_{i,i+m} = a_{i+m,i} = a, \quad i = 1, \dots, n - m, \\ & a_{ij} = 0, \text{ otherwise} \end{aligned}$	
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5-0.9 ■ <0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured					

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lyche-numerical-linear-algebra- E09679 (now)	241	—	$\begin{bmatrix} \vdots & \vdots & \cdots & \vdots \\ a_{r_2, c_1} & a_{r_2, c_1+1} & \cdots & a_{r_2, c_2} \end{bmatrix}$	$\begin{bmatrix} \vdots & \vdots & \cdots & \vdots \\ a_{r_2, c_1} & a_{r_2, c_1+1} & \cdots & a_{r_2, c_2} \end{bmatrix}$	—
basis: inferred / ink					
lyche-numerical-linear-algebra- E09796 (was)	253	■ 0.001	$\begin{array}{l} \boldsymbol{T} \boldsymbol{V} + \boldsymbol{V} \boldsymbol{T} \\ \boldsymbol{V}^{\boldsymbol{V} = \boldsymbol{S} \boldsymbol{X} \boldsymbol{S}} \boldsymbol{T} \boldsymbol{S} \boldsymbol{X} \boldsymbol{S} + \boldsymbol{S} \boldsymbol{X} \boldsymbol{S} \boldsymbol{T} \\ \stackrel{\boldsymbol{S}() \boldsymbol{S}}{\Leftrightarrow} \boldsymbol{S} \boldsymbol{T} \boldsymbol{S} \boldsymbol{X} \boldsymbol{S}^2 + \boldsymbol{S}^2 \boldsymbol{X} \boldsymbol{S} \boldsymbol{T} \boldsymbol{S} \\ \boldsymbol{S}^2 \boldsymbol{D} \\ \stackrel{\boldsymbol{T}}{\Leftrightarrow} \boldsymbol{S} \boldsymbol{X} \boldsymbol{S}^2 + \boldsymbol{S}^2 \boldsymbol{X} \boldsymbol{S}^2 \boldsymbol{D} \\ \boldsymbol{S}^2 \\ \Leftrightarrow \boldsymbol{I} / (2h) \end{array}$	$\begin{array}{l} \boldsymbol{T} \boldsymbol{V} + \boldsymbol{V} \boldsymbol{T} = h^2 \boldsymbol{F} \\ \boldsymbol{V}^{\boldsymbol{V} = \boldsymbol{S} \boldsymbol{X} \boldsymbol{S}} \boldsymbol{T} \boldsymbol{S} \boldsymbol{X} \boldsymbol{S} + \boldsymbol{S} \boldsymbol{X} \boldsymbol{S} \boldsymbol{T} = h^2 \boldsymbol{F} \\ \stackrel{\boldsymbol{S}() \boldsymbol{S}}{\Leftrightarrow} \boldsymbol{S} \boldsymbol{T} \boldsymbol{S} \boldsymbol{X} \boldsymbol{S}^2 + \boldsymbol{S}^2 \boldsymbol{X} \boldsymbol{S} \boldsymbol{T} \boldsymbol{S} = h^2 \boldsymbol{S} \boldsymbol{F} \boldsymbol{S} = h^2 \boldsymbol{G} \\ \stackrel{\boldsymbol{T}}{\Leftrightarrow} \boldsymbol{S}^2 \boldsymbol{D} \quad \boldsymbol{S} \boldsymbol{X} \boldsymbol{S}^2 + \boldsymbol{S}^2 \boldsymbol{X} \boldsymbol{S}^2 \boldsymbol{D} \\ \boldsymbol{S}^2 = h^2 \boldsymbol{G} \\ \Leftrightarrow \boldsymbol{I} / (2h) \end{array}$	—
lyche-numerical-linear-algebra- E09796 (now)	253	—	$\begin{array}{l} \boldsymbol{T} \boldsymbol{V} + \boldsymbol{V} \boldsymbol{T} \\ \boldsymbol{V}^{\boldsymbol{V} = \boldsymbol{S} \boldsymbol{X} \boldsymbol{S}} \boldsymbol{T} \boldsymbol{S} \boldsymbol{X} \boldsymbol{S} + \boldsymbol{S} \boldsymbol{X} \boldsymbol{S} \boldsymbol{T} \\ \stackrel{\boldsymbol{S}() \boldsymbol{S}}{\Leftrightarrow} \boldsymbol{S} \boldsymbol{T} \boldsymbol{S} \boldsymbol{X} \boldsymbol{S}^2 + \boldsymbol{S}^2 \boldsymbol{X} \boldsymbol{S} \boldsymbol{T} \boldsymbol{S} \\ \stackrel{\boldsymbol{T} \boldsymbol{S} = \boldsymbol{S} \boldsymbol{D}}{\Leftrightarrow} \boldsymbol{S}^2 \boldsymbol{D} \boldsymbol{X} \boldsymbol{S}^2 + \boldsymbol{S}^2 \boldsymbol{X} \boldsymbol{S}^2 \boldsymbol{D} \\ \stackrel{\boldsymbol{S}^2 = \boldsymbol{I} / (2h)}{\Leftrightarrow} \boldsymbol{D} \boldsymbol{X} + \boldsymbol{X} \boldsymbol{D} = 4h^4 \boldsymbol{G}. \end{array}$	$\begin{array}{l} \boldsymbol{T} \boldsymbol{V} + \boldsymbol{V} \boldsymbol{T} = h^2 \boldsymbol{F} \\ \stackrel{\boldsymbol{V} = \boldsymbol{S} \boldsymbol{X} \boldsymbol{S}}{\Leftrightarrow} \boldsymbol{T} \boldsymbol{S} \boldsymbol{X} \boldsymbol{S} + \boldsymbol{S} \boldsymbol{X} \boldsymbol{S} \boldsymbol{T} = h^2 \boldsymbol{F} \\ \stackrel{\boldsymbol{S}() \boldsymbol{S}}{\Leftrightarrow} \boldsymbol{S} \boldsymbol{T} \boldsymbol{S} \boldsymbol{X} \boldsymbol{S}^2 + \boldsymbol{S}^2 \boldsymbol{X} \boldsymbol{S} \boldsymbol{T} \boldsymbol{S} = h^2 \boldsymbol{S} \boldsymbol{F} \boldsymbol{S} = h^2 \boldsymbol{G} \\ \stackrel{\boldsymbol{T} \boldsymbol{S} = \boldsymbol{S} \boldsymbol{D}}{\Leftrightarrow} \boldsymbol{S}^2 \boldsymbol{D} \boldsymbol{X} \boldsymbol{S}^2 + \boldsymbol{S}^2 \boldsymbol{X} \boldsymbol{S}^2 \boldsymbol{D} = h^2 \boldsymbol{G} \\ \stackrel{\boldsymbol{S}^2 = \boldsymbol{I} / (2h)}{\Leftrightarrow} \boldsymbol{D} \boldsymbol{X} + \boldsymbol{X} \boldsymbol{D} = 4h^4 \boldsymbol{G}. \end{array}$	—
basis: inferred / ink					
lyche-numerical-linear-algebra- E09729 (was)	258	■ 0.002	$\begin{array}{l} \left(\boldsymbol{F}_{2m} \boldsymbol{P}_{2m} \right)_{p, q} \\ \left(\boldsymbol{F}_{2m} \boldsymbol{P}_{2m} \right)_{p+m, q} \\ \left(\boldsymbol{F}_{2m} \boldsymbol{P}_{2m} \right)_{p, q+m} \\ \left(\boldsymbol{F}_{2m} \boldsymbol{P}_{2m} \right)_{p+m, q+m} \end{array}$	$\begin{array}{l} \left(\boldsymbol{F}_{2m} \boldsymbol{P}_{2m} \right)_{p, q} = \omega_{2m}^{j(2k)} = \omega_m^{jk}, \\ \left(\boldsymbol{F}_{2m} \boldsymbol{P}_{2m} \right)_{p+m, q} = \omega_{2m}^{(j+m)(2k)} = \omega_m^{(j+m)k}, \\ \left(\boldsymbol{F}_{2m} \boldsymbol{P}_{2m} \right)_{p, q+m} = \omega_{2m}^{j(2k+1)} = \omega_{2m}^j \omega_m^{jk}, \\ \left(\boldsymbol{F}_{2m} \boldsymbol{P}_{2m} \right)_{p+m, q+m} = \omega_{2m}^{j+m} \left(2k + 1 \right) = \omega_{2m}^{j+m} \omega_m^{j+m} \left(2k + 1 \right) \end{array}$	—
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
lyche-numerical-linear-algebra- E00729 (now)	258	—	$(DX + XD)_{j,k} = \sum_{\ell=1}^m d_{j,\ell} x_{\ell,k} + \sum_{\ell=1}^m x_{j,\ell} d_{\ell,k} = \lambda_j x_{j,k} + \lambda_k x_{j,k} \quad \text{for all } j, k$	$(DX + XD)_{j,k} = \sum_{\ell=1}^m d_{j,\ell} x_{\ell,k} + \sum_{\ell=1}^m x_{j,\ell} d_{\ell,k} = \lambda_j x_{j,k} + \lambda_k x_{j,k} \quad \text{for all } j, k$	—
basis: inferred / ink					
lyche-numerical-linear-algebra- E00763 (was)	271	■ 0.402	$\boldsymbol{G}_{\omega} = \left[\begin{array}{ll} 1 & 0 \\ 0 & 1 \end{array} \right] - \left[\begin{array}{ll} \omega/2 & 0 \\ \omega^2/4 & \omega/2 \end{array} \right] \left[\begin{array}{ll} 2 & -1 \\ -1 & 2 \end{array} \right] = \left[\begin{array}{ll} 1-\omega & \omega/2 \\ \omega(1-\omega)/2 & 1-\omega+\omega^2/4 \end{array} \right].$	$\boldsymbol{G}_{\omega} = \left[\begin{array}{ll} l & 0 \\ 0 & 1 \end{array} \right] - \left[\begin{array}{ll} \omega/2 & 0 \\ \omega^2/4 & \omega/2 \end{array} \right] \left[\begin{array}{ll} 2 & -1 \\ -1 & 2 \end{array} \right] = \left[\begin{array}{ll} 1-\omega & \omega/2 \\ \omega(1-\omega)/2 & 1-\omega+\omega^2/4 \end{array} \right].$	—
lyche-numerical-linear-algebra- E00763 (now)	271	—	$\langle j_1, j_2 \mid j_c \rangle \quad \begin{bmatrix} \cdot & \cdot & \cdots & \cdot \\ a_{i_r, j_1} & a_{i_r, j_2} & \cdots & a_{i_r, j_c} \end{bmatrix}$	$\langle j_1, j_2 \mid j_c \rangle \quad \begin{bmatrix} \cdot & \cdot & \cdots & \cdot \\ a_{i_r, j_1} & a_{i_r, j_2} & \cdots & a_{i_r, j_c} \end{bmatrix}$	—
basis: inferred / ink					
Conf. MathPix confidence: ■ ≥ 0.9 ■ 0.5–0.9 ■ < 0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured					

Unresolved (5)

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
lyche-numerical-linear-algebra EQ0195	082	—	<pre>\begin{figure} \captionsetup{labelformat=empty} \caption{Listing 3.3 cforwardsolve} \[N_{\text{L U}}:= \frac{2}{3} n^3-\frac{1}{2} n^2-\frac{1}{6} n \]</pre>	(not rendered)	$N_{LU} := \frac{2}{3}n^3 - \frac{1}{2}n^2 - \frac{1}{6}n$
lyche-numerical-linear-algebra EQ0232	093	—	<pre>\begin{figure} \captionsetup{labelformat=empty} \caption{Listing 3.5 cbacksolve} \[\boldsymbol{A}((k+1): n,(k+1): n) \boldsymbol{b}_{\textit{k}}((k+1): n)=- \boldsymbol{A}((k+1): n, k) \textit{b}_{\textit{k}}(k), \quad k=1, \ldots, n-1 . \]</pre>	(not rendered)	$\boldsymbol{A}((k+1):n,(k+1):n)\boldsymbol{b}_k((k+1):n) = -\boldsymbol{A}((k+1):n,k)b_k(k), \quad k = 1, \dots, n-1.$
lyche-numerical-linear-algebra EQ0258	106	—	<pre>\begin{figure} \captionsetup{labelformat=empty} \caption{Listing 4.2 bandcholesky} \[0< \left(\alpha \boldsymbol{e}_{\textit{i}}+\beta \boldsymbol{e}_{\textit{j}}\right)^* \boldsymbol{A} \left(\alpha \boldsymbol{e}_{\textit{i}}+\beta \boldsymbol{e}_{\textit{j}}\right) = \alpha ^2 a_{\textit{ii}}+ \beta ^2 a_{\textit{jj}}+2 \operatorname{Re} \left(\overline{\alpha } \beta a_{\textit{ij}}\right) . \]</pre>	(not rendered)	$0 < (\leq)(\alpha \boldsymbol{e}_i + \beta \boldsymbol{e}_j)^* \boldsymbol{A}(\alpha \boldsymbol{e}_i + \beta \boldsymbol{e}_j) = \alpha ^2 a_{ii} + \beta ^2 a_{jj} + 2\operatorname{Re}(\overline{\alpha} \beta a_{ij}).$
lyche-numerical-linear-algebra EQ0855	296	—	<pre>\begin{array}{lll} \boldsymbol{t}_{\textit{0}}=\left[\right. \\ \begin{array}{c} 3 \\\ -3 / 2 \end{array} \left. \right], \\ \quad \alpha_{\textit{0}}=1 / 2, \quad \boldsymbol{x}_{\textit{1}}=\left[\right. \\ \begin{array}{c} -1 / 4 \\\ -1 / 2 \end{array} \left. \right], \\ \quad \boldsymbol{r}_{\textit{1}}=\left[\begin{array}{c} 0 \\ 3 / 4 \end{array} \right], \quad \beta_{\textit{0}}=1 / 4, \\ \boldsymbol{p}_{\textit{1}}=\left[\begin{array}{c} 3 / 8 \\ 9 / 8 \end{array} \right], \quad \boldsymbol{t}_{\textit{1}}=\left[\begin{array}{c} 0 \\ 9 / 8 \end{array} \right], \quad \alpha_{\textit{1}}=2 / 3, \quad \boldsymbol{x}_{\textit{2}}=\boldsymbol{0}, \quad \boldsymbol{r}_{\textit{2}}=\boldsymbol{0}. \\ \boldsymbol{p}_{\textit{1}}=\left[\begin{array}{c} 3 / 8 \\ 9 / 8 \end{array} \right], \quad \boldsymbol{t}_{\textit{1}}=\left[\begin{array}{c} 0 \\ 9 / 8 \end{array} \right], \quad \alpha_{\textit{1}}=2 / 3, \quad \boldsymbol{x}_{\textit{2}}=\boldsymbol{0}, \\ \quad \boldsymbol{r}_{\textit{2}}=\boldsymbol{0} . \end{array}</pre>	(not rendered)	$\boldsymbol{t}_0 = \begin{bmatrix} -\frac{3}{2} \end{bmatrix}, \quad \alpha_0 = 1/2, \quad \boldsymbol{x}_1 = \begin{bmatrix} -1/4 \\ -1/2 \end{bmatrix}, \quad \boldsymbol{r}_1 = \begin{bmatrix} 0 \\ 3/4 \end{bmatrix}, \quad \beta_0 = 1/4, \\ \boldsymbol{p}_1 = \begin{bmatrix} 3/8 \\ 3/4 \end{bmatrix}, \quad \boldsymbol{t}_1 = \begin{bmatrix} 0 \\ 9/8 \end{bmatrix}, \quad \alpha_1 = 2/3, \quad \boldsymbol{x}_2 = \boldsymbol{0}, \quad \boldsymbol{r}_2 = \boldsymbol{0}.$
lyche-numerical-linear-algebra EQ1043	353	—	<pre>\begin{figure} \[\boldsymbol{A}=\left[\right. \\ \begin{array}{llll} x & x & x & x \\ 0 & x & x & x \\ 0 & 0 & x & x \\ 0 & 0 & 0 & x \end{array} \right] \xrightarrow{\boldsymbol{P}_{12}^*} \left[\right. \\ \begin{array}{llll} \boldsymbol{x} & x & x & x \\ 0 & 0 & x & x \\ 0 & 0 & 0 & x \end{array} \right] \xrightarrow{\boldsymbol{P}_{23}^*} \left[\right. \\ \begin{array}{llll} x & x & x & x \\ x & \boldsymbol{x} & x & x \\ 0 & \boldsymbol{x} & x & x \\ 0 & 0 & 0 & x \end{array} \right] \xrightarrow{\boldsymbol{P}_{34}^*} \left[\right. \\ \begin{array}{llll} x & x & x & x \\ x & x & x & x \\ 0 & x & x & x \\ 0 & 0 & \boldsymbol{x} & x \end{array} \right] . \]</pre>	(not rendered)	$\boldsymbol{A} = \begin{bmatrix} x & x & x & x \\ 0 & x & x & x \\ 0 & 0 & x & x \\ 0 & 0 & 0 & x \end{bmatrix} \xrightarrow{\boldsymbol{P}_{12}^*} \begin{bmatrix} \boldsymbol{x} & x & x & x \\ 0 & 0 & x & x \\ 0 & 0 & 0 & x \end{bmatrix} \xrightarrow{\boldsymbol{P}_{23}^*} \begin{bmatrix} x & x & x & x \\ x & \boldsymbol{x} & x & x \\ 0 & \boldsymbol{x} & x & x \\ 0 & 0 & 0 & x \end{bmatrix} \xrightarrow{\boldsymbol{P}_{34}^*} \begin{bmatrix} x & x & x & x \\ x & x & x & x \\ 0 & x & x & x \\ 0 & 0 & \boldsymbol{x} & x \end{bmatrix}.$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured					

Flagged, not acted on — 1 of 239 shown

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
lyche-numerical-linear-algebra-E00485	191	0.917	$\begin{aligned} & \ \boldsymbol{A} \boldsymbol{B}\ = \max_{\boldsymbol{x} \neq \mathbf{0}} \frac{\ \boldsymbol{A} \boldsymbol{B} \boldsymbol{x}\ }{\ \boldsymbol{x}\ } = \max_{\boldsymbol{B} \boldsymbol{x} \neq \mathbf{0}} \frac{\ \boldsymbol{A} \boldsymbol{B} \boldsymbol{x}\ }{\ \boldsymbol{x}\ } = \max_{\boldsymbol{B} \boldsymbol{x} \neq \mathbf{0}} \frac{\ \boldsymbol{A} \boldsymbol{B} \boldsymbol{x}\ }{\ \boldsymbol{B} \boldsymbol{x}\ } \frac{\ \boldsymbol{B} \boldsymbol{x}\ }{\ \boldsymbol{x}\ } \\ & \leq \max_{\boldsymbol{y} \neq \mathbf{0}} \frac{\ \boldsymbol{A} \boldsymbol{y}\ }{\ \boldsymbol{y}\ } \max_{\boldsymbol{x} \neq \mathbf{0}} \frac{\ \boldsymbol{B} \boldsymbol{x}\ }{\ \boldsymbol{x}\ } = \ \boldsymbol{A}\ \ \boldsymbol{B}\ \end{aligned}$	$\ \boldsymbol{A} \boldsymbol{B}\ = \max_{\boldsymbol{x} \neq \mathbf{0}} \frac{\ \boldsymbol{A} \boldsymbol{B} \boldsymbol{x}\ }{\ \boldsymbol{x}\ } = \max_{\boldsymbol{B} \boldsymbol{x} \neq \mathbf{0}} \frac{\ \boldsymbol{A} \boldsymbol{B} \boldsymbol{x}\ }{\ \boldsymbol{x}\ } = \max_{\boldsymbol{B} \boldsymbol{x} \neq \mathbf{0}} \frac{\ \boldsymbol{A} \boldsymbol{B} \boldsymbol{x}\ }{\ \boldsymbol{B} \boldsymbol{x}\ } \frac{\ \boldsymbol{B} \boldsymbol{x}\ }{\ \boldsymbol{x}\ } \\ \leq \max_{\boldsymbol{y} \neq \mathbf{0}} \frac{\ \boldsymbol{A} \boldsymbol{y}\ }{\ \boldsymbol{y}\ } \max_{\boldsymbol{x} \neq \mathbf{0}} \frac{\ \boldsymbol{B} \boldsymbol{x}\ }{\ \boldsymbol{x}\ } = \ \boldsymbol{A}\ \ \boldsymbol{B}\ .$	
Conf. MathPix confidence: ≥ 0.9 0.5-0.9 < 0.5 — no value Residual render vs scan (inkdrill): C component W weak S stable N noise K clean not measured					

The remaining 238 flagged rows are stated as a count.

Each differs from its scan by fewer than 20 components, which is the tail of the distribution rather than its bulk: 38 are component (C), 200 weak (W). By MathPix confidence: 172 at ≥ 0.9 , 41 in 0.6-0.9, 25 below 0.6, 0 with none. Out/465 sampled twenty rows drawn from the ≥ 0.9 band corpus-wide and found eighteen purely typographic differences, one typographic plus a crop overrun, one borderline and *no* unambiguous content errors — so a high-confidence flag here is evidence of a rendering difference, not of a defect. Every row is in `report.ink.json`.